

Dose-Weighted Sobel Metrics and Structure-Tensor-Based Edge Orientation for Proton Therapy Complexity Analysis

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Abstract

We define a set of heterogeneity metrics based on the 3-D Sobel edge filter applied to CT Hounsfield-unit volumes in the neighbourhood of the Bragg peak. Two extensions over a plain gradient-magnitude summary are introduced. First, every voxel’s contribution is weighted by the local dose, so that tissue interfaces outside the irradiated region are discounted. Second, rather than discarding the directional information carried by the gradient vector, we aggregate gradient directions via the *dose-weighted structure tensor*, whose eigendecomposition yields the dominant edge normal and a set of interpretable scalar shape descriptors. These descriptors are intended as predictors of model error in machine-learning-based proton dose prediction. In this document they are positioned as candidate difficulty estimators inside an *active beamlet mining* pipeline that selects beamlet-induced CT patches for expensive Monte Carlo dose calculation (Section 1).

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1 Research Framework: Active Beamlet Mining

The Sobel and structure-tensor metrics developed in the remainder of this document are intended as a building block of a larger active-learning pipeline for proton dose calculation. Before introducing the technical material we state the broader research formulation in which these metrics will be embedded; the dose-weighted Sobel descriptors defined in Section 4 onward instantiate the abstract *difficulty estimator* that appears in the formulation below.

1.1 Clean Research Formulation: Active Beamlet Mining

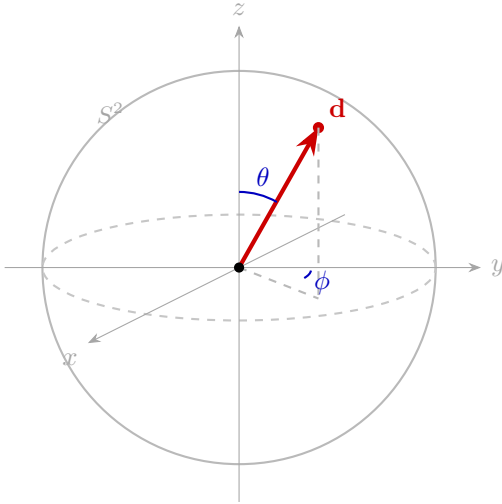
Let $I \in \mathbb{R}^{D \times H \times W}$ denote a 3D CT image (depth-first axis ordering, consistent with Section 2) and let M_θ denote a trained, but imperfect, deep-learning-based dose calculation model. The objective is to improve M_θ by actively selecting beamlet-induced CT patches for which expensive Monte Carlo dose calculation is expected to provide the highest training value.

A proton beamlet is parameterized as

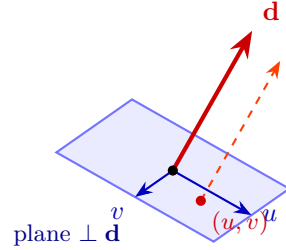
$$b = (\mathbf{d}, u, v),$$

where $\mathbf{d} \in S^2$ is the beam direction and (u, v) defines the lateral beamlet position in the plane orthogonal to $\mathbf{d}$. The space of feasible beamlets for a given CT image is denoted by

$$\mathcal{B}(I).$$



(a) $\mathbf{d} \in S^2$: a direction in 3-D



(b) Beamlet $b = (\mathbf{d}, u, v)$

Figure 1: Two views of the proton-beamlet parametrisation. **(a)** The unit 2-sphere $S^2 = \{\mathbf{d} \in \mathbb{R}^3 : \|\mathbf{d}\|_2 = 1\}$ is the space of beam directions: every point on its surface is one unit vector $\mathbf{d}$ from the origin, and conversely every direction in 3-D space is one such point. Two angles parametrise $\mathbf{d}$ — the polar angle $\theta \in [0, \pi]$ from the z -axis and the azimuthal angle $\phi \in [0, 2\pi)$ in the xy -plane — confirming that $\mathbf{d}$ has two degrees of freedom. **(b)** Once $\mathbf{d}$ is chosen, the lateral position (u, v) fixes the pencil beam within the plane orthogonal to $\mathbf{d}$ (blue parallelogram). The dashed orange ray is the actual pencil beam: a line parallel to $\mathbf{d}$ passing through the offset point (u, v) . Total degrees of freedom: $2 + 2 = 4$ per beamlet.

Given a beamlet $b \in \mathcal{B}(I)$, a raytracing operator extracts a beam-aligned CT patch

$$x_b = R(I, b),$$

where R denotes the patch extraction operator. The patch x_b represents the local anatomical region through which the proton beamlet travels and is expressed in *beamlet coordinates*: axis 0 is the depth direction along $\mathbf{d}$, and axes 1, 2 lie in the lateral plane orthogonal to $\mathbf{d}$. All Sobel and structure-tensor metrics introduced in Sections 2–6 operate on x_b in this beamlet-aligned frame.

Assume that a difficulty estimator

$$\Delta(x_b)$$

is available. This estimator assigns a scalar score to each extracted patch, reflecting the expected difficulty of the patch for the current dose calculation model M_θ .

The active learning problem is then formulated as the selection of a subset of beamlets

$$S^\star = \{b_1, \dots, b_K\} \subset \mathcal{B}(I)$$

whose corresponding patches should be evaluated with Monte Carlo simulation and added to the training database.

A naive strategy would select the beamlets with the largest difficulty scores:

$$S^\star = \arg \max_{S \subset \mathcal{B}(I), |S|=K} \sum_{b \in S} \Delta(x_b).$$

However, this may result in selecting many redundant beamlets that traverse similar anatomical regions.

To control redundancy and promote variability, we introduce a patch embedding function

$$E : \mathcal{X} \rightarrow \mathbb{R}^m,$$

which maps each beamlet-induced patch x_b to a low-dimensional latent representation

$$z_b = E(x_b).$$

The set

$$\mathcal{M}_I = \{E(R(I, b)) : b \in \mathcal{B}(I)\}$$

defines a CT-specific beamlet-patch manifold. This manifold represents the geometry of all candidate beamlet-induced patches for the given CT image.

Let $\mathcal{D}_{\text{train}}$ denote the current training database, containing previously sampled beamlet patches. The novelty of a candidate beamlet b with respect to the existing training data is defined as

$$N(b) = \min_{q \in \mathcal{D}_{\text{train}}} \|E(x_b) - E(x_q)\|_2.$$

For a partially selected set S we additionally need a measure of how similar a candidate beamlet b is to those already chosen. Let $z_b = E(x_b) = E(R(I, b))$ denote the latent code of patch x_b . We use the Gaussian (RBF) similarity *kernel* between latent codes,

$$k(z, z') = \exp\left(-\frac{\|z - z'\|_2^2}{\sigma^2}\right) \in [0, 1],$$

where $\sigma > 0$ is the kernel *bandwidth*. This is the same RBF kernel used in support-vector machines and Gaussian-process regression; σ is *not* the standard deviation of a probability distribution but a length-scale parameter in the embedding space, controlling how rapidly the similarity score decays with distance: $k = 1$ at zero distance, $k \approx 0.37$ at distance σ , and $k \rightarrow 0$ as the distance grows large.

Why a bounded kernel for redundancy but a raw distance for novelty? The novelty term $N(b)$ only has to be *monotone* in distance — greater distance from the training database is always better — so the raw L_2 distance suffices. The redundancy term, by contrast, enters the acquisition function with a fixed weight λ alongside $\Delta(x_b)$ and $N(b)$, each of which has its own intrinsic scale. If redundancy were the raw squared distance to the nearest already-selected beamlet its magnitude would depend on the (arbitrary) scale of the embedding E , and λ would have to be re-tuned every time E changes. Mapping distance through k bounds the redundancy contribution to $[0, 1]$ and makes λ directly interpretable as the weight assigned to a perfect duplicate.

Set-level objective. The principled active beamlet mining objective is to maximise total per-beamlet utility while penalising pairwise similarity within the selected set:

$$S^* = \arg \max_{S \subset \mathcal{B}(I), |S|=K} \sum_{b \in S} [\Delta(x_b) + \gamma N(b)] - \lambda \sum_{\{b_i, b_j\} \subset S} k(z_{b_i}, z_{b_j}),$$

where $\sum_{\{b_i, b_j\} \subset S}$ denotes a sum over *unordered pairs* of distinct beamlets in S , and $\gamma \geq 0$, $\lambda \geq 0$ trade off novelty and diversity against difficulty. This set-level form is the defining objective: it is the standard Gaussian-kernel diversification criterion and, with appropriate normalisation, connects to determinantal-point-process active learning.

Greedy approximation. Solving the set-level problem exactly is combinatorial. We use a greedy approximation: starting from $S_0 = \emptyset$, at step t we add the candidate that maximises a per-element acquisition function,

$$A(b; S_{t-1}) = \Delta(x_b) + \gamma N(b) - \lambda R_{\text{red}}(b, S_{t-1}),$$

where the redundancy contribution is taken to be the kernel similarity of b to its closest already-selected neighbour,

$$R_{\text{red}}(b, S) = \max_{b_j \in S} k(z_b, z_{b_j}).$$

The greedy step is then

$$b_t^* = \arg \max_{b \in \mathcal{B}(I) \setminus S_{t-1}} A(b; S_{t-1}), \quad S_t = S_{t-1} \cup \{b_t^*\},$$

iterated until $|S_t| = K$.

Relation between the set-level objective and the greedy surrogate. The exact marginal contribution of adding b to S_{t-1} under the set-level pairwise penalty is the *sum* $\sum_{b_j \in S_{t-1}} k(z_b, z_{b_j})$. The max-version used in R_{red} replaces this sum with its single largest summand and is therefore a stricter, more conservative diversity heuristic: a candidate is penalised by its closest neighbour in S alone, so a near-duplicate of any one already-selected beamlet is rejected even when far from all the others. The max form is the default because it is cheaper to evaluate ($O(|S_{t-1}|)$ per candidate) and produces well-spread selections in practice; the corresponding sum-version surrogate $R_{\text{red}}^\Sigma(b, S) = \sum_{b_j \in S} k(z_b, z_{b_j})$, which is the exact greedy marginal of the set-level objective, will be evaluated as an alternative.

The selected beamlets define the set of patches

$$\mathcal{X}_{\text{MC}} = \{x_b = R(I, b) : b \in S^*\},$$

for which Monte Carlo dose calculations are performed. The resulting labeled samples are added to the training database:

$$\mathcal{D}_{\text{train}}^{\text{new}} = \mathcal{D}_{\text{train}} \cup \{(x_b, D_b^{\text{MC}}) : b \in S^*\}.$$

The dose calculation model is then updated by minimizing the supervised training objective

$$\theta^* = \arg \min_{\theta} \sum_{(x, D^{\text{MC}}) \in \mathcal{D}_{\text{train}}^{\text{new}}} \mathcal{L}(M_{\theta}(x), D^{\text{MC}}).$$

This formulation defines active beamlet mining as a geometry-aware acquisition problem over the space of feasible proton beamlets. The manifold representation is not constructed over the CT image itself, but over the set of beamlet-induced patches generated from the CT image:

$$\mathcal{M}_I = \{E(R(I, b)) : b \in \mathcal{B}(I)\}.$$

The acquisition process therefore prioritizes beamlets that are simultaneously difficult for the current model, novel with respect to the existing training database, and non-redundant with respect to already selected beamlets.

1.2 Connection to the Remainder of This Document

The remainder of this document develops one concrete realisation of the difficulty estimator $\Delta(x_b)$ defined above. For each beamlet-induced patch $x_b = R(I, b)$ a vector of dose-weighted Sobel and structure-tensor descriptors,

$$\phi(x_b) = (\bar{S}_{dw}(x_b), A(x_b), \theta(x_b), E_{\text{total}}(x_b)) \in \mathbb{R}^4,$$

is computed (Sections 4–6), each component isolating a distinct, physically interpretable aspect of the local tissue geometry traversed by the beamlet. The difficulty estimator is then any scalar function of these descriptors,

$$\Delta(x_b) = h(\phi(x_b)).$$

Three reasonable forms of h will be considered:

1. a *single descriptor*, e.g. $\theta(x_b)$ alone, motivated by its direct link to range uncertainty (Section 6.2);
2. a *hand-tuned weighted sum* of components of ϕ chosen on physical grounds; or
3. a *regression model* (e.g. a small MLP, random forest, or regularised linear model) fitted to predict the observed gamma pass rate from ϕ across the existing training set.

The empirical comparison of Method DW and Method TH (Section 9) supplies the evidence required to choose between weighted and threshold-masked variants of each descriptor before the form of h is finalised.

The patch embedding $E(x_b)$ used for novelty $N(b)$ and redundancy $R_{\text{red}}(b, S)$ is logically distinct from $\Delta(x_b)$ and may use a different representation: a learned encoder (for example the bottleneck of M_{θ} itself), a pre-trained 3-D CNN, or the descriptor vector $\phi(x_b)$ itself. This document does not commit to a specific embedding; that choice is deferred to a separate methodological note.

2 Notation and Setup

Let $I(\mathbf{x})$ denote the CT intensity (Hounsfield units) at voxel $\mathbf{x} = (z, y, x) \in \mathbb{Z}^3$, and let $d(\mathbf{x}) \geq 0$ denote the ground-truth absorbed dose at the same voxel. **Notational convention.** In Section 1 the symbol I denotes the full patient CT image; here and in subsequent sections we re-use I for the *beam-aligned patch* $x_b = R(\cdot, b)$ produced by the raytracing operator, with the beamlet index b suppressed for notational economy. With this convention axis 0 (z) is the depth direction along the beam, and axes 1, 2 (y, x) span the lateral plane orthogonal to it. Physical

coordinates are obtained by multiplying voxel indices by the isotropic voxel spacing $(\delta_z, \delta_y, \delta_x)$ in mm.

The *Bragg-peak location* $\mathbf{x}_{BP}$ is the voxel that maximises the depth-integrated dose in the axial direction:

$$\mathbf{x}_{BP} = \arg \max_{\mathbf{x}} d(\mathbf{x}). \quad (1)$$

The *analysis sphere* of radius R (mm) centred on $\mathbf{x}_{BP}$ is

$$\Omega_R = \left\{ \mathbf{x} \in \mathbb{Z}^3 : (z - z_{BP})^2 \delta_z^2 + (y - y_{BP})^2 \delta_y^2 + (x - x_{BP})^2 \delta_x^2 \leq R^2 \right\}. \quad (2)$$

3 Three-Dimensional Sobel Gradient

The 3-D Sobel operator approximates the partial derivative of I along each axis by convolving with axis-specific kernels. The resulting gradient vector at voxel $\mathbf{x}$ is

$$\mathbf{g}(\mathbf{x}) = \begin{pmatrix} G_z(\mathbf{x}) \\ G_y(\mathbf{x}) \\ G_x(\mathbf{x}) \end{pmatrix}, \quad \|\mathbf{g}(\mathbf{x})\| = \sqrt{G_z^2 + G_y^2 + G_x^2}. \quad (3)$$

The unit-norm gradient, wherever $\|\mathbf{g}\| > 0$, is the *edge normal*:

$$\hat{n}(\mathbf{x}) = \frac{\mathbf{g}(\mathbf{x})}{\|\mathbf{g}(\mathbf{x})\|}. \quad (4)$$

Why gradient direction matters

The Sobel gradient magnitude $\|\mathbf{g}\|$ captures only *how strong* a tissue interface is, not *how it is oriented*. In proton therapy, the range of a pencil beam is sensitive to the material it traverses along the beam axis (here $\hat{z}$). An interface whose normal is parallel to $\hat{z}$ (i.e. the interface is perpendicular to the beam) perturbs the water-equivalent path length and therefore the Bragg-peak depth. An interface whose normal is orthogonal to $\hat{z}$ (i.e. the interface runs along the beam) causes lateral heterogeneity but less range shift. Capturing the dominant edge orientation is therefore physically motivated.

4 Dose-Weighted Sobel Metrics

4.1 Motivation

The existing implementation aggregates $\|\mathbf{g}\|$ uniformly over Ω_R , treating every voxel equally regardless of whether it receives dose. A tissue interface far from any isodose surface does not affect dose complexity, so its gradient should not inflate the heterogeneity score. We replace uniform aggregation with dose weighting.

4.2 Dose-Weighted Mean Gradient Magnitude

$$\bar{S}_{dw} = \frac{\sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) \|\mathbf{g}(\mathbf{x})\|}{\sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x})} \quad (5)$$

This is a proper weighted average of edge strength, where the weights are the local dose values. Voxels with $d(\mathbf{x}) = 0$ contribute nothing; voxels near the dose maximum contribute most.

The metric is defined for $\sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) > 0$; if the sphere contains no dose it defaults to zero.

4.3 Relationship to the Existing Metric

The current unweighted mean corresponds to setting $d(\mathbf{x}) = 1$ for all $\mathbf{x} \in \Omega_R$:

$$\bar{S}_{unweighted} = \frac{1}{|\Omega_R|} \sum_{\mathbf{x} \in \Omega_R} \|\mathbf{g}(\mathbf{x})\|. \quad (6)$$

Equation (5) is a strict generalisation.

4.4 Method TH: Threshold-Masked Mean

The second approach replaces continuous dose weighting with a binary mask defined by a global dose threshold. Let $d_{\text{grid}} = \max_{\mathbf{x} \in \mathcal{V}} d(\mathbf{x})$ be the maximum dose over the *entire CT grid* $\mathcal{V}$ (not just inside the sphere). The active set at threshold $\varepsilon = 0.05$ is:

$$\Omega_R^{TH} = \{\mathbf{x} \in \Omega_R : d(\mathbf{x}) \geq \varepsilon d_{\text{grid}}\}. \quad (7)$$

The threshold metric is the unweighted mean of gradient magnitudes over this active set:

$$\boxed{\bar{S}_{TH} = \frac{1}{|\Omega_R^{TH}|} \sum_{\mathbf{x} \in \Omega_R^{TH}} \|\mathbf{g}(\mathbf{x})\|.} \quad (8)$$

All voxels inside Ω_R^{TH} contribute equally; those outside (below 5% of the global dose peak) contribute nothing. The same binary mask is applied to the structure tensor:

$$\mathbf{J}_{TH} = \sum_{\mathbf{x} \in \Omega_R^{TH}} \mathbf{g}(\mathbf{x}) \otimes \mathbf{g}(\mathbf{x}), \quad (9)$$

from which anisotropy A_{TH} , beam-edge angle θ_{TH} , and total edge energy E_{TH} are derived analogously to Sections 6.2–15.

Why use the global grid maximum? Normalising the threshold to the whole-grid dose peak (rather than the in-sphere peak) makes the threshold consistent across samples and physically grounded: $0.05 d_{\text{grid}}$ corresponds to the clinical 5% isodose surface of the plan, a standard boundary used in radiotherapy dosimetry to separate irradiated from unirradiated tissue.

Key difference from Method DW. Method DW (5) assigns each voxel a weight proportional to its dose — a soft, continuous transition. Method TH (8) makes a hard binary decision: a voxel is either “in” or “out”. Within the active set, every voxel contributes equally regardless of whether it sits at the dose peak or just above the threshold.

5 Dose-Weighted Structure Tensor

5.1 The Cancellation Problem: Why We Cannot Average Gradient Vectors

The Sobel operator gives us, at every voxel, an arrow whose direction is perpendicular to the nearest tissue boundary. A first idea would be to simply average all these arrows over the sphere to find the “dominant” interface direction. Unfortunately, this approach fails by construction.

Consider a thin bone slab passing through the centre of the sphere (Figure 2). The slab creates two parallel interfaces: a left face (soft tissue on the left, bone on the right) and a right face (bone on the left, soft tissue on the right). The Sobel gradient points from lower to higher CT intensity, i.e. from soft tissue toward bone:

- At the *left face*, the adjacent soft tissue is to the left of the bone, so the gradient in that tissue region points **rightward** (toward the denser bone).
- At the *right face*, the adjacent soft tissue is to the right, so the gradient in that tissue region points **leftward** (again toward the bone, but now from the opposite side).

Both gradient sets are equally strong and point in exactly opposite directions; their sum over the sphere is zero (Figure 2a). Yet the interface is as real and as sharp as possible. In clinical practice this scenario — a bone plate (e.g. a rib or the iliac crest) running across the beam path — is one of the most relevant sources of range uncertainty in proton therapy.

This is not a numerical quirk; it is an unavoidable consequence of the fact that a gradient vector is a *signed* quantity. Any method that sums signed vectors will exhibit this cancellation whenever the sphere contains an interface that is symmetric about its centre, which is precisely the situation we care about most.

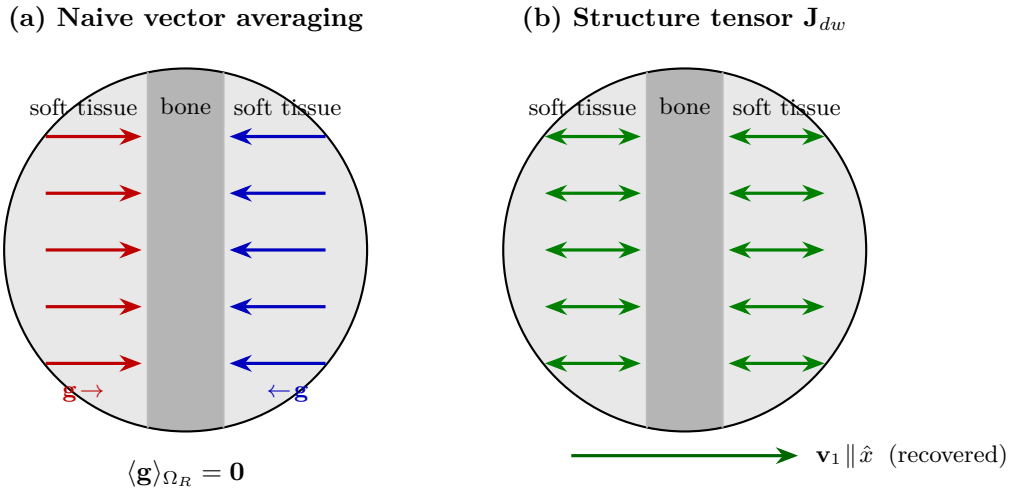


Figure 2: Illustration of the gradient-vector cancellation problem and its resolution via the structure tensor. Both panels show a 2-D cross-section of the analysis sphere (circle); a thin bone slab (dark grey) crosses the centre vertically, with soft tissue (light grey) on both sides. **(a)** Red arrows: the Sobel gradient in the left soft-tissue region points rightward into the bone (higher HU). Blue arrows: the gradient in the right soft-tissue region points leftward, also toward the bone. Both sets are equal in magnitude but opposite in direction; their mean over Ω_R is exactly zero, even though a strong, well-defined interface is present. **(b)** Replacing each gradient vector $\mathbf{g}$ with its outer product $\mathbf{g} \otimes \mathbf{g}$ removes the sign: contributions from both faces become identical (double-headed arrows). Eigendecomposition of the accumulated $\mathbf{J}_{dw}$ correctly recovers $\mathbf{v}_1 \parallel \hat{x}$ as the dominant edge normal.

5.2 The Outer-Product Trick

The solution is to represent each gradient not as a signed arrow but as a *direction preference* — a quantity that is the same for $\mathbf{g}$ and $-\mathbf{g}$.

The mathematical object with this property is the *outer product* $\mathbf{g} \otimes \mathbf{g}$, a 3×3 matrix:

$$\mathbf{g} \otimes \mathbf{g} = \begin{pmatrix} G_z \\ G_y \\ G_x \end{pmatrix} \begin{pmatrix} G_z & G_y & G_x \end{pmatrix} = \begin{pmatrix} G_z^2 & G_z G_y & G_z G_x \\ G_y G_z & G_y^2 & G_y G_x \\ G_x G_z & G_x G_y & G_x^2 \end{pmatrix}. \quad (10)$$

The crucial observation is that squaring removes the sign: $(-\mathbf{g}) \otimes (-\mathbf{g}) = \mathbf{g} \otimes \mathbf{g}$. Left and right

arrows — the two faces of the same boundary — now produce *identical* matrices and therefore reinforce rather than cancel when summed.

Intuition. Think of it as a voting system. Each voxel “votes” for the axis along which its gradient is most strongly oriented. The vote is not “I point left” or “I point right” (which would cancel), but rather “the variation here is along the x -direction” — a statement that is symmetric and thus consistent for both faces of the same interface. Squaring the gradient components is exactly what turns a signed arrow into this unsigned, direction-only vote.

5.3 The Dose-Weighted Structure Tensor

Summing the dose-weighted outer products over the analysis sphere gives the *dose-weighted structure tensor*:

$$\mathbf{J}_{dw} = \sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) \mathbf{g}(\mathbf{x}) \otimes \mathbf{g}(\mathbf{x}) = \sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) \begin{pmatrix} G_z^2 & G_z G_y & G_z G_x \\ G_y G_z & G_y^2 & G_y G_x \\ G_x G_z & G_x G_y & G_x^2 \end{pmatrix}. \quad (11)$$

$\mathbf{J}_{dw}$ is a 3×3 real symmetric matrix with non-negative diagonal entries. It is the dose-weighted accumulation of all gradient direction preferences inside the sphere: every voxel contributes proportionally to both its local dose and the squared strength of its edge.

What does $\mathbf{J}_{dw}$ encode? The diagonal entry $[\mathbf{J}_{dw}]_{zz} = \sum d(\mathbf{x}) G_z^2$ records how much dose-weighted gradient energy is oriented along the z -axis. Similarly for yy and xx . The off-diagonal entry $[\mathbf{J}_{dw}]_{zy} = \sum d(\mathbf{x}) G_z G_y$ records correlations between the z - and y -components of the gradient: a large positive value means gradients tend to be oriented in a direction that is simultaneously in z and in y , i.e. diagonally between the two axes. The full 3×3 structure encodes the dominant orientation as a *shape in direction space*, not just a single number.

5.4 Eigendecomposition: Reading the Shape of the Tensor

A symmetric positive semi-definite matrix such as $\mathbf{J}_{dw}$ can be thought of as defining an *ellipsoid* in three-dimensional space. The axes of the ellipsoid are the *eigenvectors*; their lengths are the square roots of the *eigenvalues*.

$$\mathbf{J}_{dw} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T, \quad \mathbf{\Lambda} = \text{diag}(\lambda_1, \lambda_2, \lambda_3), \quad \lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0. \quad (12)$$

The columns of $\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$ are three mutually orthogonal unit vectors — the principal directions of the gradient distribution.

Geometric picture. Imagine plotting one gradient arrow per voxel (each scaled by $\sqrt{d(\mathbf{x})}$) as a point cloud in 3-D direction space. The structure tensor captures the “shape” of this cloud:

- If the cloud is elongated like a sausage — almost all arrows point along one direction — the ellipsoid is cigar-shaped: one long axis (λ_1 large) and two short axes (λ_2, λ_3 small). This corresponds to a single dominant planar interface.
- If the cloud is spread equally in all directions — a sphere — all three eigenvalues are equal. This corresponds to edges pointing in every direction, e.g. a spherical bone inclusion.
- Intermediate cases (e.g. a cylindrical structure) produce a disc-like ellipsoid: one small axis (λ_3) and two large ones ($\lambda_1 \approx \lambda_2$).

Why does the long axis give the edge normal? At a sharp planar interface the gradient is large only in the direction *perpendicular* to the surface (because the CT intensity changes most rapidly across the boundary). All arrows therefore accumulate along this perpendicular direction: the cloud is elongated exactly in the direction of the interface normal. The long axis of the ellipsoid — eigenvector $\mathbf{v}_1$ — is the dominant edge normal.

Interpretation of eigenvectors and eigenvalues

$\mathbf{v}_1, \lambda_1$: The *dominant edge normal* — the direction perpendicular to the most prominent tissue interface in the irradiated part of the sphere. A large λ_1 means the corresponding interface is strong and well-localised.

$\mathbf{v}_3, \lambda_3$: The direction *along* the dominant edge (the tangential direction of least gradient variation). A small λ_3 relative to λ_1 confirms that the dominant interface is nearly planar.

$\mathbf{v}_2, \lambda_2$: The remaining principal direction, orthogonal to both. Together, $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ form a complete orthonormal frame that describes the full 3-D geometry of the gradient distribution.

6 Scalar Descriptors from the Structure Tensor

The structure tensor is a 3×3 matrix — too rich to use directly as a regression feature or a comparison metric. We extract three scalars that each capture a distinct, physically meaningful aspect of the edge geometry.

6.1 Edge Anisotropy

$$A = \frac{\lambda_1 - \lambda_3}{\lambda_1 + \lambda_2 + \lambda_3} \in [0, 1] \quad (13)$$

Intuition. Imagine comparing two objects: a rugby ball and a soccer ball. The rugby ball has one long axis and two short ones — it is anisotropic. The soccer ball looks the same from every direction — it is isotropic. The anisotropy A does exactly this comparison for the ellipsoid described by the structure tensor.

$A \approx 1$ (**cigar-shaped ellipsoid**): $\lambda_1 \gg \lambda_2, \lambda_3$. Almost all dose-weighted gradient energy is concentrated along a single direction. This indicates one dominant, well-defined planar interface — for example, a flat bone plate crossing the sphere. The dominant edge normal $\mathbf{v}_1$ is a reliable and meaningful descriptor in this case.

$A \approx 0$ (**spherical ellipsoid**): $\lambda_1 \approx \lambda_2 \approx \lambda_3$. Gradient energy is spread uniformly across all directions. This indicates that the sphere contains many interfaces oriented in all directions (e.g. trabecular bone, or a spherical heterogeneity). There is no single dominant edge, and $\mathbf{v}_1$ loses its meaning as a “dominant normal” — any direction would be equally valid.

A high anisotropy means both that the dominant edge is geometrically well-defined *and* that the angle θ (Section 6.2) is a trustworthy number. When A is low, θ should be treated with caution.

6.2 Dominant-Edge-to-Beam Angle

The proton beam travels along the z -axis, $\hat{z} = (1, 0, 0)^T$ in the (z, y, x) convention used here. The angle between the dominant edge normal $\mathbf{v}_1$ and the beam is:

$$\theta = \arccos(|\mathbf{v}_1 \cdot \hat{z}|) \in [0^\circ, 90^\circ] \quad (14)$$

The absolute value is necessary because the sign of an eigenvector is arbitrary: $\mathbf{v}_1$ and $-\mathbf{v}_1$ describe the same geometric direction, but would give dot products of opposite sign without the absolute value.

Intuition: the wall and the ball. Imagine throwing a ball along a corridor and asking how much a wall inside the corridor slows it down.

- If the wall is placed *across* the corridor (perpendicular to the ball’s path), the ball hits the wall straight on — maximum interaction.
- If the wall runs *along* the corridor (parallel to the ball’s path), the ball flies past it with minimal disturbance.

The angle θ is exactly this: it measures how much the dominant tissue interface is “across the corridor” of the proton beam.

Physical interpretation. In proton therapy, the energy deposited by a pencil beam at depth depends on the *water-equivalent path length* (WEPL) through the materials along the beam axis. A tissue interface changes the WEPL only insofar as it *crosses* the beam path.

$\theta \approx 0^\circ$ (**interface perpendicular to beam**): The edge normal is parallel to $\hat{z}$. The beam must pass through the interface. Each proton traverses a step change in density, shifting the Bragg-peak depth. This is the configuration of maximum range uncertainty and is most likely to degrade model prediction accuracy.

$\theta \approx 90^\circ$ (**interface parallel to beam**): The edge normal is perpendicular to $\hat{z}$. The interface runs alongside the beam without being crossed. The WEPL and therefore the Bragg-peak depth are less perturbed; any residual effect is on the lateral dose profile rather than the range.

6.3 Total Edge Energy

$$E_{total} = \text{tr}(\mathbf{J}_{dw}) = \lambda_1 + \lambda_2 + \lambda_3 = \sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) \|\mathbf{g}(\mathbf{x})\|^2 \quad (15)$$

The trace of the structure tensor equals the dose-weighted sum of squared gradient magnitudes. This identity follows directly from the definition (11): the diagonal entries of $\mathbf{g} \otimes \mathbf{g}$ are G_z^2 , G_y^2 , G_x^2 , whose sum is $\|\mathbf{g}\|^2$.

E_{total} is the dose-weighted analogue of the existing `sum_sobel_bp` metric (which uses unsquared magnitudes and no dose weighting). Its advantage is that it decomposes cleanly into the three eigenvalues, making it consistent with the rest of the structure-tensor framework.

Relationship between E_{total} and A . The two metrics are complementary. E_{total} tells you *how much* total edge energy is present in the irradiated region; A tells you *how concentrated* that energy is along one direction. A sphere with a single sharp bone plate and a sphere filled with fine trabecular bone could have similar E_{total} , but very different A : the bone plate gives $A \approx 1$, the trabecular bone gives $A \approx 0$.

Table 1: Proposed dose-weighted Sobel metrics. All quantities are computed inside the analysis sphere Ω_R centred on the Bragg peak.

Symbol	Equation	Physical meaning
$\bar{S}_{dw}$	(5)	Dose-weighted mean Sobel gradient magnitude. Overall edge intensity in the irradiated region.
$\mathbf{J}_{dw}$	(11)	Dose-weighted structure tensor (3×3 matrix).
$\mathbf{v}_1$	eigenvector of $\mathbf{J}_{dw}$, λ_1	Dominant edge normal direction (3-D unit vector).
A	(13)	Edge anisotropy: how planar vs. omnidirectional the dominant interface is.
θ	(14)	Angle between dominant edge normal and beam axis; directly related to range-uncertainty risk.
E_{total}	(15)	Dose-weighted total edge energy (trace of $\mathbf{J}_{dw}$).

7 Summary of Proposed Metrics

8 Implementation Notes

1. **Dose volume.** At training/analysis time, the ground-truth dose $d(\mathbf{x})$ is used for weighting. When the metrics are applied post-hoc to model predictions, the predicted dose could be substituted; the choice should be documented per experiment.
2. **Sphere extraction.** The current implementation computes Sobel on the full CT volume and then masks. This remains correct for the structure tensor; only the accumulation step changes.
3. **Numerical stability.** If $\sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x}) < \varepsilon_{tol}$ (no dose in sphere), all metrics default to zero and $\mathbf{v}_1$ is undefined. The anisotropy A should also be set to zero in this case.
4. **Eigendecomposition.** $\mathbf{J}_{dw}$ is a 3×3 real symmetric matrix. `numpy.linalg.eigh` provides an efficient, numerically stable decomposition with eigenvalues guaranteed to be real and returned in ascending order (reverse to get descending $\lambda_1 \geq \lambda_2 \geq \lambda_3$).
5. **Sign convention for $\mathbf{v}_1$.** The eigenvector sign is arbitrary. The beam-edge angle (14) uses the absolute value of the dot product, making it sign-invariant. If the vector itself is stored or visualised, a canonical sign convention should be chosen (e.g. $v_{1,z} \geq 0$).

9 Comparison of Method DW and Method TH

9.1 What Each Method Assumes

Before designing experiments it is worth stating explicitly what each method assumes about the relationship between dose and edge relevance.

Method DW assumes that *the relevance of a tissue interface scales continuously with the local dose*. A voxel receiving twice the dose of a neighbour contributes twice as much to the metric. This is a soft, graded view: there is no sharp boundary between “relevant” and “irrelevant”; every dose-positive voxel matters to some degree.

Method TH assumes that *there is a hard clinical boundary between irradiated and unirradiated tissue*, defined by the 5% global isodose surface. Once a voxel is classified as irradiated, it is treated as equally relevant regardless of its actual dose value. This is a binary view that aligns with standard radiotherapy dose prescription concepts.

The two assumptions lead to different behaviours in specific geometric configurations, summarised in Table 2.

Table 2: Expected behaviour of Method DW and Method TH in representative geometric configurations.

Scenario	Method DW	Method TH
Interface well inside the high-dose region	Up-weighted; contributes strongly	Included; contributes equally
Interface near the 5% isodose surface	Down-weighted by low dose; contributes little	Either fully included or fully excluded
All dose in the sphere is uniform (flat dose plateau)	Equivalent to unweighted mean over sphere	Equivalent to unweighted mean over thresholded sphere
Highly peaked dose (narrow Bragg peak)	Strongly up-weights voxels near the dose maximum	Treats all above-threshold voxels equally

9.2 Metrics to Be Compared

For every sample in the training set, both methods are computed for all scalar descriptors. Table 3 lists the full set of paired quantities.

Table 3: Scalar descriptors computed under each method. All quantities are extracted from the analysis sphere Ω_R .

Descriptor	Method DW	Method TH
Mean gradient magnitude	$\bar{S}_{DW}$ (5)	$\bar{S}_{TH}$ (8)
Edge anisotropy	A_{DW}	A_{TH}
Beam-edge angle	θ_{DW}	θ_{TH}
Total edge energy	E_{DW}	E_{TH}

9.3 Comparison Protocol

The comparison is structured around three complementary analyses.

9.3.1 Correlation with Model Error

For each descriptor and each method, compute the Spearman rank correlation ρ with the model prediction error, measured by:

- Gamma pass rate (GPR) at 3%/3 mm — primary endpoint;
- RMSE and MAPE — secondary endpoints.

Because we have eight descriptor-method combinations (4×2), we report all correlations in a single table, comparing ρ_{DW} and ρ_{TH} side by side for each descriptor. A statistically significant difference in $|\rho|$ between the two methods for the same descriptor indicates that the weighting strategy matters for that descriptor.

9.3.2 Ranking Consistency

Even if both methods have similar correlation with model error, they may disagree on which individual samples are the most heterogeneous. To quantify this, compute Kendall’s rank correlation coefficient τ between the DW and TH rankings for each descriptor:

$$\tau = \frac{\# \text{ concordant pairs} - \# \text{ discordant pairs}}{\frac{1}{2}N(N-1)}, \quad (16)$$

where N is the number of samples.

$\tau \approx 1$ means the two methods produce nearly identical rankings; $\tau \approx 0$ means they are essentially independent. Low τ for a given descriptor signals that the weighting choice is consequential and that one method must be selecting genuinely different information than the other.

9.3.3 Discordance Analysis

Identify the $k = 10$ samples with the largest absolute rank difference $|\text{rank}_{DW} - \text{rank}_{TH}|$ for the primary descriptor $\bar{S}$. For each discordant sample, examine:

1. The dose distribution inside Ω_R : is it narrowly peaked (Bragg peak well inside the sphere) or broadly spread?
2. The active set size $|\Omega_R^{TH}|$ relative to $|\Omega_R|$: what fraction of the sphere falls above the global 5 % threshold?
3. Whether the tissue interface responsible for the high Sobel magnitude overlaps with the high-dose zone or lies in the periphery of the sphere.

This analysis reveals the geometric conditions under which the two methods diverge and provides a physical interpretation of the discordance.

9.4 Expected Sources of Discordance

Two structural differences between the methods generate predictable discordances:

Dose-peak localisation. When the Bragg peak is sharp and sits near the centre of the sphere, the dose is highly concentrated over a small number of voxels. Method DW up-weights those voxels strongly. If the dominant interface happens to be at the dose peak, Method DW gives it disproportionately high weight relative to Method TH. Conversely, if the dominant interface is at the sphere boundary where dose is low, Method DW suppresses it while Method TH may still include it (provided the dose there exceeds 5 % of the global maximum).

Global vs. local threshold reference. Method TH’s threshold is anchored to the global dose maximum d_{grid} , which is consistent across samples but may include or exclude very different fractions of the sphere depending on the absolute dose level. A low-energy beam (small d_{grid}) may have most of the sphere above threshold; a high-energy beam with a narrow Bragg peak may have very few voxels above it. Method DW is immune to this variation because it normalises by the local dose sum $\sum_{\mathbf{x} \in \Omega_R} d(\mathbf{x})$.

9.5 Decision Criterion

The preferred method will be selected based on the following ordered criteria:

1. **Predictive power:** the method with higher $|\rho|$ with GPR for the majority of descriptors is preferred.

2. **Consistency:** in the case of comparable predictive power, prefer the method with more stable behaviour across the discordant samples identified in Section [9.3.3](#).
3. **Interpretability:** Method TH has a direct clinical interpretation (5% isodose surface); Method DW has a cleaner mathematical formulation. This criterion is used as a tiebreaker.